HDOC Day-29 - Solution

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Problem / Objective

Perform menu-driven array operations: count even, odd, positive, negative and zero elements; search for an element using linear search; merge two arrays; and exit.

Algorithm

Read the working array once. In Count, traverse it once and update parity and sign counters. In Linear Search, scan from index 0 until the key matches or the array ends. In Merge, read two arrays and copy the first followed by the second into a third array. Repeat the menu until Exit is selected.

C Program

```
#include <stdio.h>
void count(int a[],int n){
int e=0,o=0,p=0,ng=0,z=0,i;
for(i=0;
i<n;
i++){
if(a[i]%2==0)e++;
else o++;
if(a[i]>0)p++;
else if(a[i]<0)ng++;
else z++;
}
printf("Even=%d Odd=%d Positive=%d Negative=%d Zero=%d\n",e,o,p,ng,z);
}

void search(int a[],int n){
int x,i;
printf("Element to search: ");
scanf("%d",&x);
for(i=0;
i<n;
i++)if(a[i]==x){
printf("Found at position %d\n",i+1);
return;
}
printf("Not found\n");
}

void merge(void){
int n,m,i;
printf("Sizes: ");
scanf("%d%d",&n,&m);
int a[n],b[m],c[n+m];
printf("First array: ");
for(i=0;
i<n;
i++){
scanf("%d",&a[i]);
c[i]=a[i];
}
printf("Second array: ");
for(i=0;
i<m;
i++){
scanf("%d",&b[i]);
c[n+i]=b[i];
}
printf("Merged: ");
for(i=0;
i<n+m;
i++)printf("%d ",c[i]);
printf("\n");
}
```

```
int main(void){
int n,i,ch;
printf("Array size: ");
scanf("%d",&n);
int a[n];
printf("Elements: ");
for(i=0;
i<n;
i++)scanf("%d",&a[i]);
do{
printf("\n1.Count 2.Linear search 3.Merge arrays 4.Exit\nChoice: ");
scanf("%d",&ch);
switch(ch){
case 1:count(a,n);
break;
case 2:search(a,n);
break;
case 3:merge();
break;
case 4:break;
default:printf("Invalid choice\n");
}
}
while(ch!=4);
return 0;
}
```

Test Cases / Expected Results

Case	Input	Expected	Status
1	[-2,-1,0,3,4]	Even=3, Odd=2, Positive=2, Negative=2, Zero=1	Pass
2	[5,8,2], key=8	Found at position 2	Pass
3	[1,2] and [3,4,5]	Merged: 1 2 3 4 5	Pass

Result / Notes

Linear search works on an unsorted array and has O(n) worst-case time complexity.